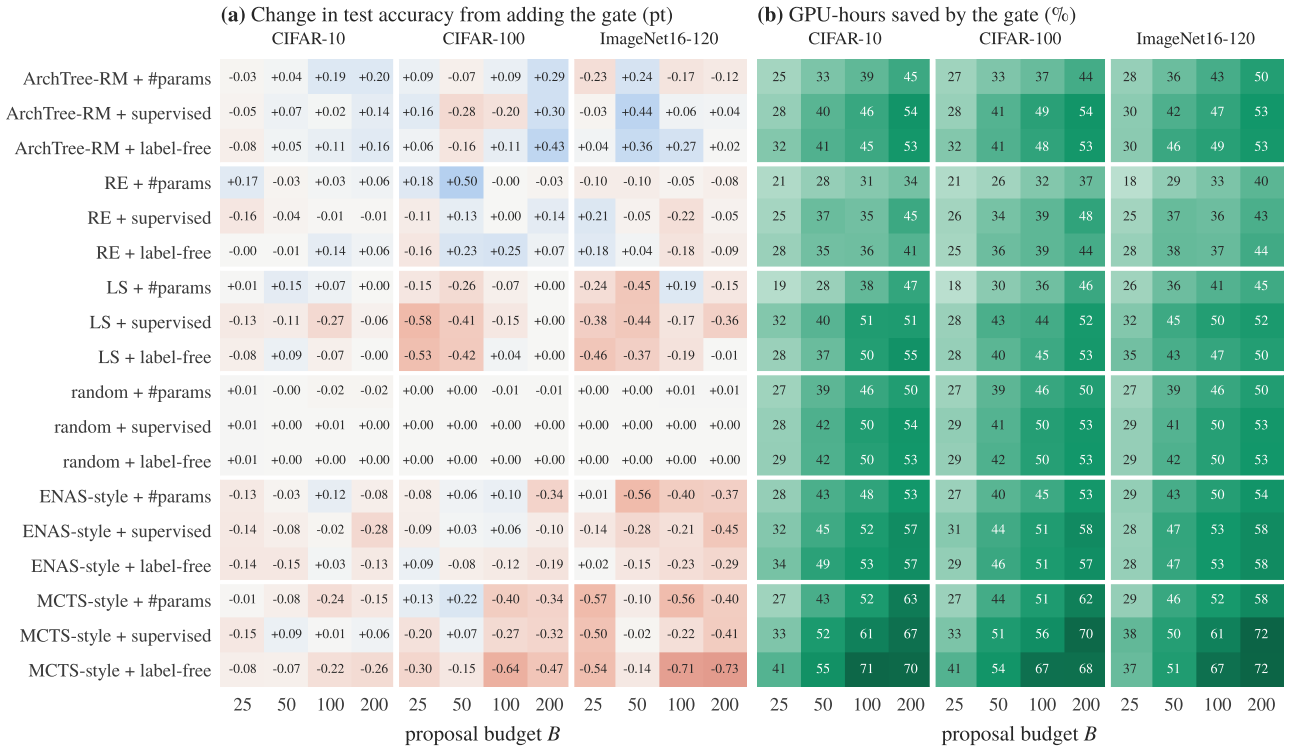

(a) Change in test accuracy from adding the gate (pt)
(b) GPU-hours saved by the gate (%)
CIFAR-10
CIFAR-100
ImageNet16-120
ArchTree-RM + #params -0.03 +0.04 +0.19 +0.20 +0.09 -0.07 +0.09 +0.29 -0.23 +0.24 -0.17 -0.12 25 33 39 45 27 33 37 44 28 36 43 50
ArchTree-RM + supervised -0.05 +0.07 +0.02 +0.14 +0.16 -0.28 -0.20 +0.30 -0.03 +0.44 +0.06 +0.04 28 40 46 54 28 41 49 54 30 42 47 53
ArchTree-RM + label-free -0.08 +0.05 +0.11 +0.16 +0.06 -0.16 +0.11 +0.43 +0.04 +0.36 +0.27 +0.02 32 41 45 53 32 41 48 53 30 46 49 53
RE + #params +0.17 -0.03 +0.03 +0.06 +0.18 +0.50 -0.00 -0.03 -0.10 -0.10 -0.05 -0.08 21 28 31 34 21 26 32 37 18 29 33 40
RE + supervised -0.16 -0.04 -0.01 -0.01 -0.11 +0.13 +0.00 +0.14 +0.21 -0.05 -0.22 -0.05 25 37 35 45 26 34 39 48 25 37 36 43
RE + label-free -0.00 -0.01 +0.14 +0.06 -0.16 +0.23 +0.25 +0.07 +0.18 +0.04 -0.18 -0.09 28 35 36 41 25 36 39 44 28 38 37 44
LS + #params +0.01 +0.15 +0.07 +0.00 -0.15 -0.26 -0.07 +0.00 -0.24 -0.45 +0.19 -0.15 19 28 38 47 18 30 36 46 26 36 41 45
LS + supervised -0.13 -0.11 -0.27 -0.06 -0.58 -0.41 -0.15 +0.00 -0.38 -0.44 -0.17 -0.36 32 40 51 51 28 43 44 52 32 45 50 52
LS + label-free -0.08 +0.09 -0.07 -0.00 -0.53 -0.42 +0.04 +0.00 -0.46 -0.37 -0.19 -0.01 28 37 50 55 28 40 45 53 35 43 47 50
random + #params +0.01 -0.00 -0.02 -0.02 +0.00 +0.00 -0.01 -0.01 +0.00 +0.00 +0.01 +0.01 27 39 46 50 27 39 46 50 27 39 46 50
random + supervised +0.01 +0.00 +0.01 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 28 42 50 54 29 41 50 53 29 41 50 53
random + label-free +0.01 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 29 42 50 53 29 42 50 53 29 42 50 53
ENAS-style + #params -0.13 -0.03 +0.12 -0.08 -0.08 +0.06 +0.10 -0.34 +0.01 -0.56 -0.40 -0.37 28 43 48 53 27 40 45 53 29 43 50 54
ENAS-style + supervised -0.14 -0.08 -0.02 -0.28 -0.09 +0.03 +0.06 -0.10 -0.14 -0.28 -0.21 -0.45 32 45 52 57 31 44 51 58 28 47 53 58
ENAS-style + label-free -0.14 -0.15 +0.03 -0.13 +0.09 -0.08 -0.12 -0.19 +0.02 -0.15 -0.23 -0.29 34 49 53 57 29 46 51 57 28 47 53 58
MCTS-style + #params -0.01 -0.08 -0.24 -0.15 +0.13 +0.22 -0.40 -0.34 -0.57 -0.10 -0.56 -0.40 27 43 52 63 27 44 51 62 29 46 52 58
MCTS-style + supervised -0.15 +0.09 +0.01 +0.06 -0.20 +0.07 -0.27 -0.32 -0.50 -0.02 -0.22 -0.41 33 52 61 67 33 51 56 70 38 50 61 72
MCTS-style + label-free -0.08 -0.07 -0.22 -0.26 -0.30 -0.15 -0.64 -0.47 -0.54 -0.14 -0.71 -0.73 41 55 71 70 41 54 67 68 37 51 67 72
25 50 100 200
proposal budget B